MIL-T-60877A (AR)
30 April 1986
SUPERSEDING
MIL-T-60877 (MU)
15 August 1969

MILITARY SPECIFICATION

TEST SET, DISTRIBUTION BOX, TS-2659/TSM-100A

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Distribution Box Test Set, hereinafter referred to as the "test set", which forms a part of the Electronics System Test Set type AN/TSM-100A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-T-43435	-	Tape: Impregnated, Lacing and Tying
MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4931

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

MIL-C-60898	-	Circuit Card Assemblies, General Specification for
MIL-C-60898/1	-	Circuit Card Assembly, Logic Function Resistors
MIL-C-60898/2	-	Circuit Card Assembly, Relay Sense Resistors
MIL-C-60898/3	-	Circuit Card Assembly, Relay Sense and K12 Operation
MIL-C-60898/4	-	Circuit Card Assembly, K6 Operation and Sequence
MIL-C-60898/5	-	Circuit Card Assembly, K8 Operation and Sequence
MIL-C-60898/6	-	Circuit Card Assembly, K11 Operation and Sequence
MIL-C-60898/7	-	Circuit Card Assembly, Relay Signal Inverters
MIL-C-60898/8	-	Circuit Card Assembly, Noise Suppression

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.3)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

F10554006	-	Panel Assembly, Distribution Box Test Unit
F10554007	-	Schematic Diagram, Electrical, Distribution Box, Test Unit

PACKAGING DATA SHEET

10554006	-	Packaging of Test Set, Distribution Box

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The test set shall be manufactured in accordance with Drawing 10554006 and drawings pertaining thereto. The subassemblies of the test set shall have been inspected and accepted in accordance with their respective drawings or specification prior to their installation into the test set.

3.2 General specification. Where applicable, the test set shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and Tolerances

b. Inorganic protective surface finishes

c. Workmanship

d. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assembly. The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modification as follows:

(1) Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.

(2) Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005 inch.

(3) Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.

(4) Drying. The drying process shall not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

b. Wiring. The requirements for wiring of MIL-C-46357 shall apply with modification as follows:

(1) Spot ties. Tape; Type 1, Finish B per MIL-T-43435 may be utilized without the need for varnish on the knots.

(2) Wire termination. Removal of insulation by controlled heat methods shall be acceptable.

(3) Chassis cleaning. Compressed air or vacuum may be utilized for cleaning. Vibration shall not be used to clean the chassis.

c. Threaded fasteners. The requirements for threaded fasteners of MIL-C-46357 shall apply with modification as follows:

(1) Thread projection. Maximum projection of screws shall be that of the next standard length screw beyond the minimum projection, unless electrical or mechanical interference exists.

d. Marking. The requirements for marking of MIL-C-46357 shall apply with modification as follows:

(1) Etching and engraving. Solvo photo etch process is acceptable.

3.2.3 Circuit card assemblies. The circuit card assemblies MIL-C-60898/1 through MIL-C-60898/8 shall meet the requirements of their respective detail sheets.

3.3 Environmental service conditions.

3.3.1 Vibration (non-operating). The test set shall meet the requirements of 3.4 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency (Hz ± 2 percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.3.2 Temperatures.

3.3.2.1 Operating. The test set shall meet the requirements of 3.4 during exposure at any temperature between +32°F and +131°F and after it is thermally stabilized at both low and high ambient temperature extremes.

3.3.2.2 Non-operating (storage). The test set shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization), and shall meet the requirements of 3.4 when returned to between +60°F and +90°F (room ambient).

3.3.3 Humidity non-operating. The test set shall meet the requiremets of 3.4 and shall not suffer any damage or deterioration which shall impair satisfactory performance, after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.4 Shock. The test set shall not suffer damage and shall meet the requirements of 3.4 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds. The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 milliseconds.

3.3.5 Altitude.

3.3.5.1 Operating. The test set shall meet the requirements of 3.4 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg). The test set shall not be damaged or adversely affected due to such exposure.

3.3.5.2 Non-operating. The test set shall meet the requirements of 3.4 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg). The test set shall not be damaged or adversely affected due to such exposure.

3.4 Electrical.

3.4.1 Electrical continuity. Continuity shall be in accordance with the Distribution Box (DB) Test Set electrical schematic Drawing 10554007. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.2 Functional.

3.4.2.1 Conditions of operation. Unless otherwise specified, the Functional requirements of 3.4.2 shall be met with the power supplies connected as shown in Figure 1 and the test system power +24VDC and +5VDC switches in the "ON" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

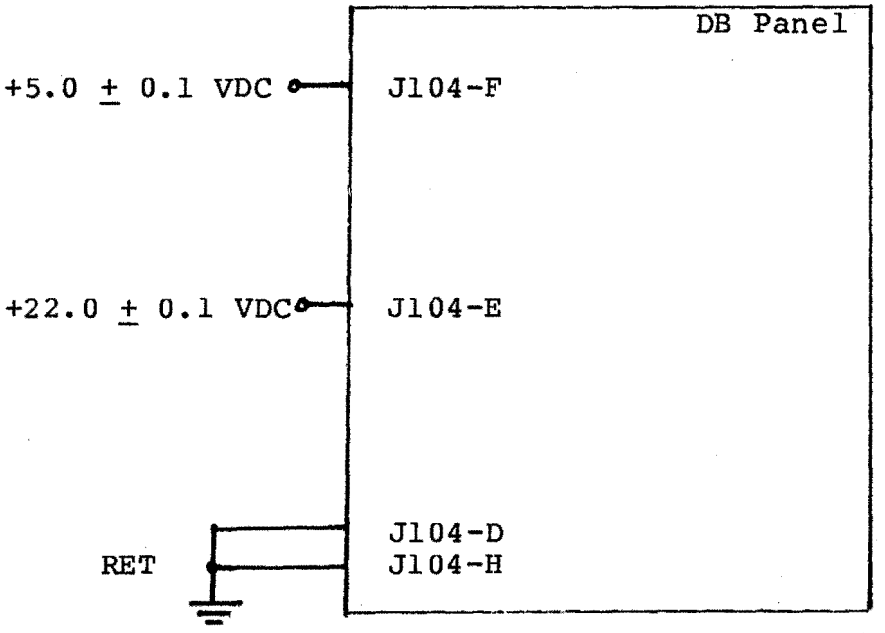


FIGURE 1. Distribution box power supply connections.

3.4.2.2 Test system power switches. The +5 VDC lamp shall light only when the +5 VDC switch is in the "ON" position. The +24 VDC lamp shall light only when both the +5 VDC and +24 VDC switches are in the "ON" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

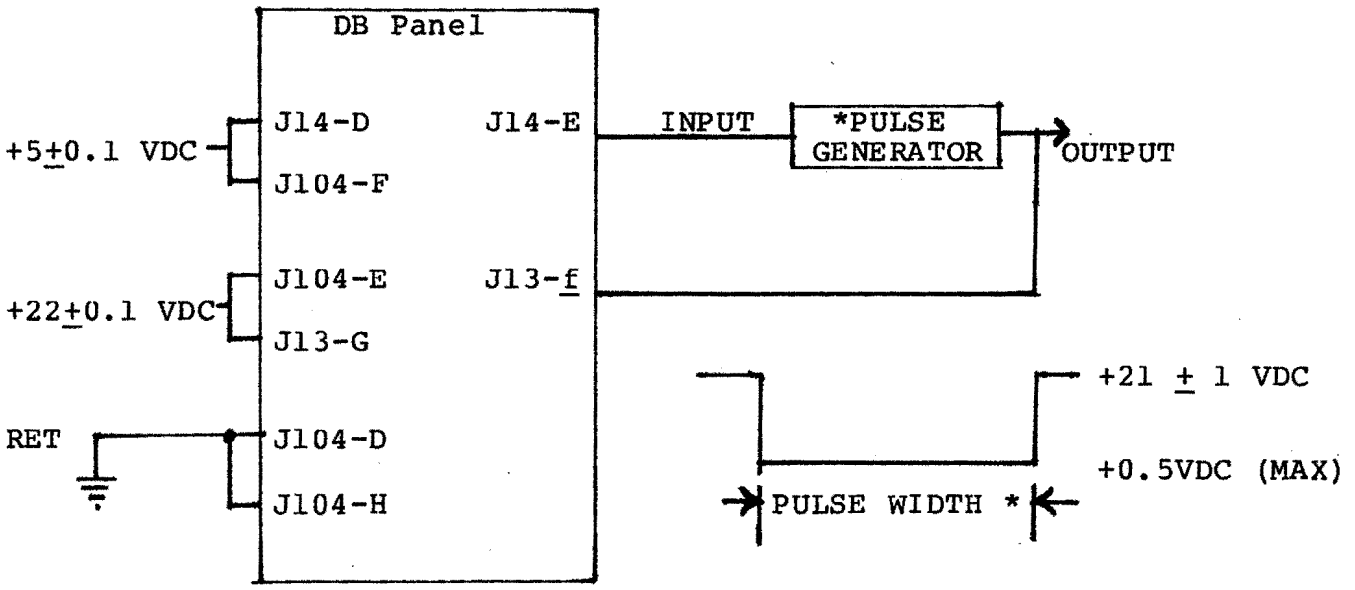


FIGURE 2. Distribution box K12 test setup.

* When the LAST ROUND switch is in the "OFF" position, the input to the pulse generator is open and the output shall be 21 ± 1 VDC. When the LAST ROUND switch is placed to the "ON" position the output of the pulse generator shall go to a maximum of +0.5 VDC for a period as specified in 3.4.2.3.a. and 3.4.2.3.f. and return to +21 ± 1 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.3 K12 Lamp delay. Connect the DB panel as shown in Figure 2 and perform the following sequence:

a. Set the pulse generator to provide a 240 ± 5 millisecond pulse.

b. Place LAST ROUND switch to the "OFF" position.

c. Depress RESET switch.

d. Place CHANNEL A and CHANNEL B switches to K12 position.

e. Place LAST ROUND switch to "ON" position, K12 lamp shall light after a delay of 240 ± 5 msec.

f. Set the pulse generator to provide a 160 ± 5 msec pulse.

g. Repeat steps b. thru e. except K12 lamp shall not light after LAST ROUND switch is placed to the "ON" position.

h. Remove +22 VDC from J13-G.

i. Repeat steps a. thru e. except K12 lamp shall not light after LAST ROUND switch is placed to the "ON" position.

3.4.2.4 K6 Lamp delay. Connect the DB panel as shown in Figure 3 and perform the following sequence:

a. Set the pulse generator to provide a 540 ± 5 msec pulse.

b. Place the BURST LIMIT switch to the "OFF" position.

c. Place LAST ROUND switch to the "OFF" position.

d. Place CHANNEL A and CHANNEL B switches to K6 position.

e. Place the TRIGGER switch to the "ON" position.

f. Depress RESET switch.

g. Place the LAST ROUND switch to the "ON" position, the K6 lamp shall light for 540 ± 5 msec.

h. Set the pulse generator to provide a 460 ± 5 msec delay.

i. Repeat steps b. thru g. except that K6 lamp shall light and not go out after the 460 ± 5 msec pulse.

j. Remove +22 VDC from J12-A.

k. Repeat steps a. thru g. except that K6 lamp shall not light after LAST ROUND switch is placed to the "ON" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

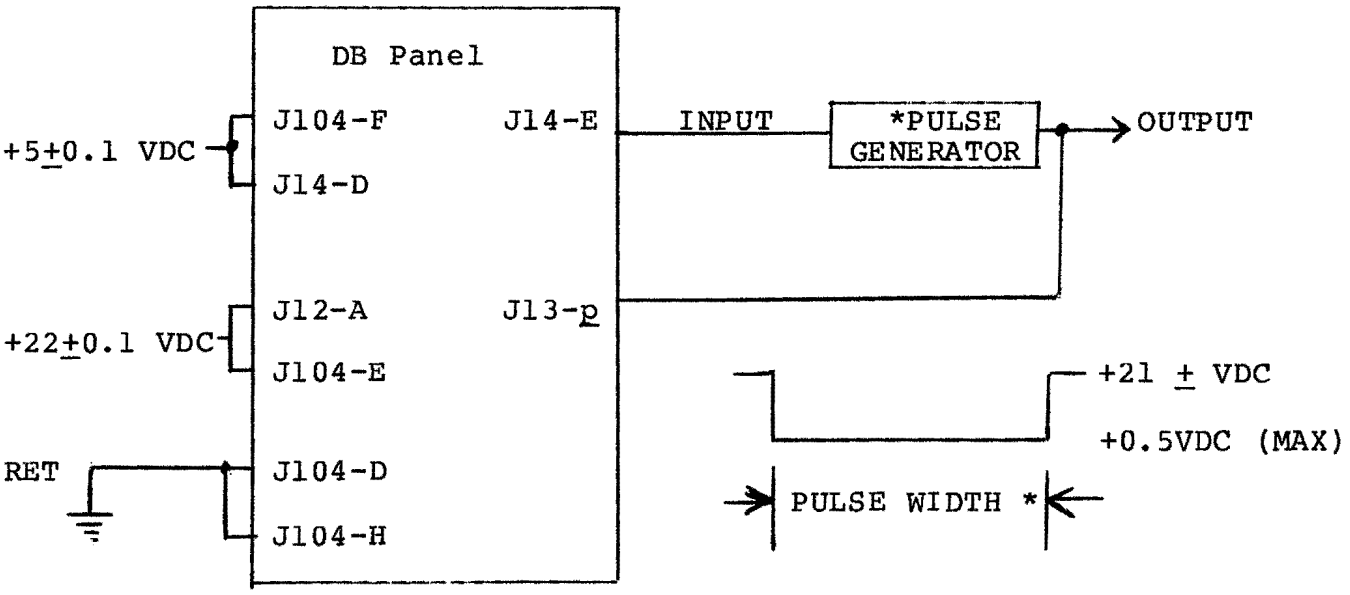


FIGURE 3. Distribution box K6 test setup.

* When the LAST ROUND switch is in the "OFF" position, the input to the pulse generator is open and the output shall be 21 ± 1 VDC. When the LAST ROUND switch is placed to the "ON" position (+5 VDC is placed on the input) the output of the pulse generator shall go to +0.5 VDC (Max.) for a period as specified in 3.4.2.4.a and 3.4.2.4.h. and return to +21 ± 1 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.5 K8 Lamp delay. Connect the DB panel as shown in Figure 4 and perform the following sequence:

a. Set the pulse generator to provide a 240 ± 5 msec delay.

b. Place the TRIGGER switch to the "OFF" position.

c. Place the RATE switch to the LO position and CHANNEL A and CHANNEL B switches to K8 position.

d. Depress RESET switch.

e. Place TRIGGER switch to "ON" position, after 240 ± 5 msec the K8 lamp shall light.

f. Set pulse generator to provide a 160 ± 5 msec delay.

g. Place TRIGGER switch to the "OFF" position.

h. Depress RESET switch, K8 lamp shall go out.

i. Place TRIGGER switch to "ON" position, K8 lamp shall not light.

j. Remove pulse generator circuit from DB panel and place +21 ± 1 VDC to J13-h.

k. Place RATE switch to HI position, K8 lamp shall light.

l. Remove +21 ± 1 VDC from J13-h, K8 lamp shall go out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

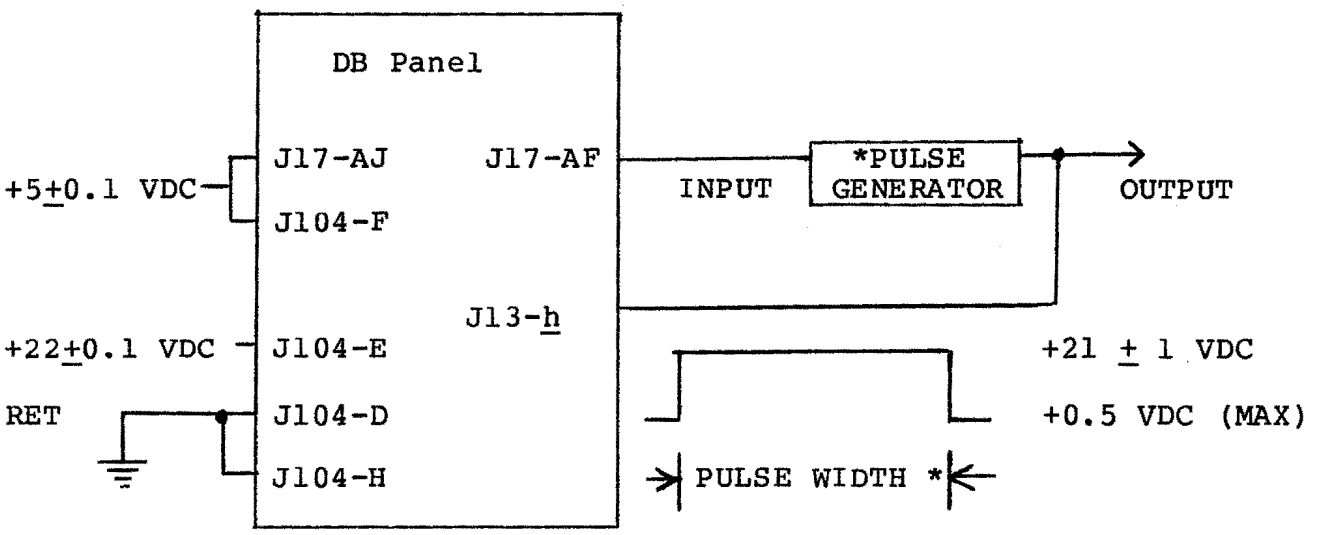


FIGURE 4. Distribution box K8 test setup.

* When the TRIGGER switch is placed in the OFF position, the input to the PULSE GENERATOR is open and the OUTPUT shall be a maximum of +0.5 VDC. When the trigger switch is placed to "ON" position +5 VDC is applied to the INPUT, the OUTPUT shall go to +21 ± 1 VDC for a period as specified in 3.4.2.5.a. and 3.4.2.5.f. and return to a maximum of +0.5 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.6 K11 Lamp delay. Connect the DB panel as shown in Figure 5 and perform the following sequence:

a. Set the pulse generator to provide a 1065 ± 5 msec delay period.

b. Place the CLEAR MODE switch to the "OFF" position.

c. Place the BURST LIMIT switch to the "ON" position.

d. Place CHANNEL A and CHANNEL B switches to K11 position.

e. Depress RESET switch.

f. Place CLEAR MODE switch to AUTO position, after 1065 ± 5 msec K11 lamp shall light.

g. Set pulse generator to provide a 935 ± 5 msec delay.

h. Place the CLEAR MODE switch to the "OFF" position and depress RESET switch, K11 lamp shall go out.

i. Place the CLEAR MODE switch to the AUTO position, K11 lamp shall not light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

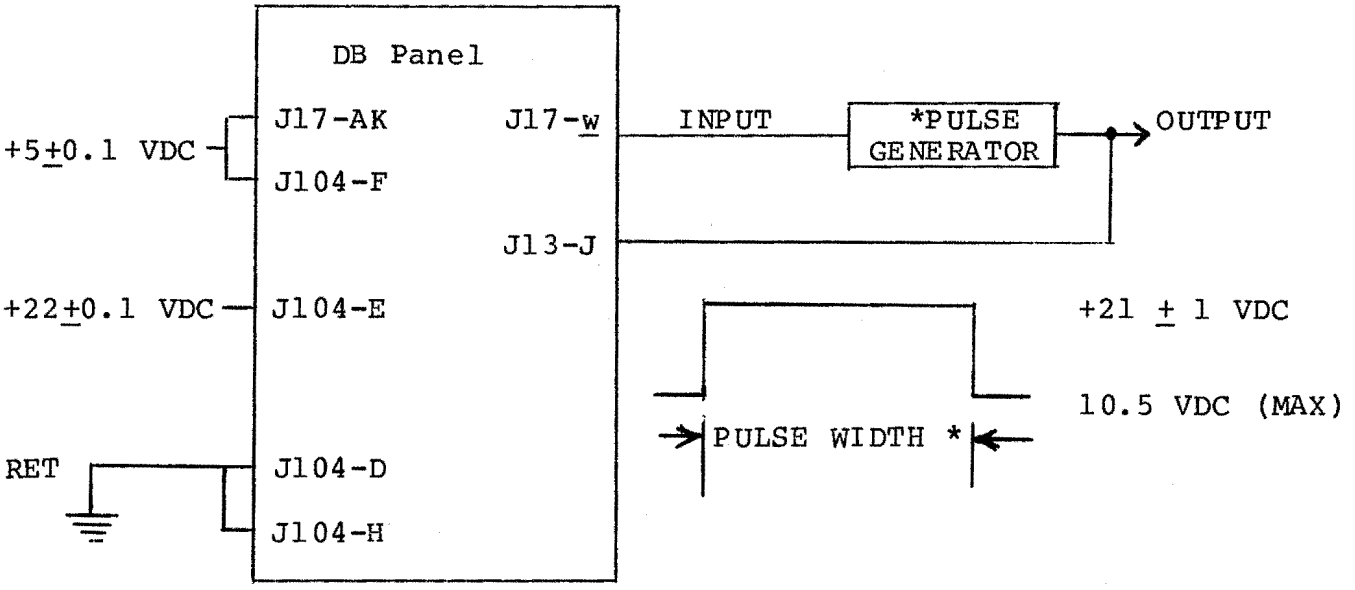


FIGURE 5. Distribution box K11 test setup.

* When the CLEAR MODE switch is placed in the "OFF" position, the INPUT to the pulse generator is open and the OUTPUT shall be a maximum of 0.5 VDC. When the CLEAR MODE switch is placed to the "AUTO" position +5 VDC is applied to the INPUT, the OUTPUT shall go to +21 ± 1 VDC for a period as specified in 3.4.2.6.a and 3.4.2.6.g. and return to a maximum of +0.5 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.7 Relay time selectors.

3.4.2.7.1 Low input. With the RELAY TIME SELECTORS CHANNEL A and CHANNEL B switches set to the position specified in Table I and +1.1 ± 0.1 VDC applied to the CONNECTOR-PINS, the voltage at INPUT A and INPUT B connectors shall be +20.5 ± 1.5 VDC and the associated INDICATOR lamps shall not light.

3.4.2.7.2 High input. With the RELAY TIME SELECTORS CHANNEL A and CHANNEL B switches set to the position specified in Table I and +20 ± 0.1 VDC applied to the CONNECTOR-PINS, the voltage at INPUT A and INPUT B connectors shall be less than +0.5 VDC and the associated INDICATOR lamp shall light except K11.

TABLE I. Relay indicator tests.

INPUT VOLTAGE & CONNECTOR-PIN	RELAY TIME SELECTORS		INDICATOR LAMP
	CHANNEL A	CHANNEL B	
J17-AE	K5	K5	K5
J13-h	K8	K8	K8-(Rate switch in HI Position)
J13-j	K10	K10	K10
J13-J	K11	K11	---
J15-B	BRAKE	BRAKE	BRAKE

3.4.2.8 K7 lamp delay. With the RELAY TIME SELECTORS CHANNEL A and CHANNEL B switches placed in the "K7" position apply the input voltages to J13-F and J17-j and monitor the output voltage at INPUT A and INPUT B connectors as outlined in Table II.

TABLE II. K7 Relay test.

INPUT (TOL. ± 0.1 VDC)		OUTPUT (TOL. ± 0.5 VDC)	
J13-F	J17-j	INPUT A	INPUT B
22	22	21.6	
OPEN	22	0.6*	
22	OPEN	21.6	
OPEN	OPEN	21.6	

*K7 indicator lamp shall light.

3.4.2.9 K9 lamp delay. With the RELAY TIME SELECTORS, CHANNEL A and CHANNEL B placed in the K9 positions apply the input voltages to J13-Q and J17-m. The output voltage at INPUT A and INPUT B connectors shall be as outlined in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE III. K9 Relay test.

INPUTS (TOL. + 0.1 VDC)		OUTPUTS (TOL. + 0.5 VDC)	
J13-Q	J17-m	INPUT A	INPUT B
22	22	21.6	
OPEN	22	0.6*	
22	OPEN	21.6	
OPEN	OPEN	21.6	

*K9 indicator lamp shall light.

3.4.2.10 Switch indicators. With the connections to the Distribution Box Test Set as shown in Figure 6, the indicator lamp specified in Table IV shall LIGHT when the associated "ON-OFF" switch is in the "ON" position and shall not be LIT when the switch is in the "OFF" position.

TABLE IV. Switch indicator tests.

SWITCH	INDICATOR
Right Limit	K15
Left Limit	K16
EL Limit	K13
ACQ Time	K14

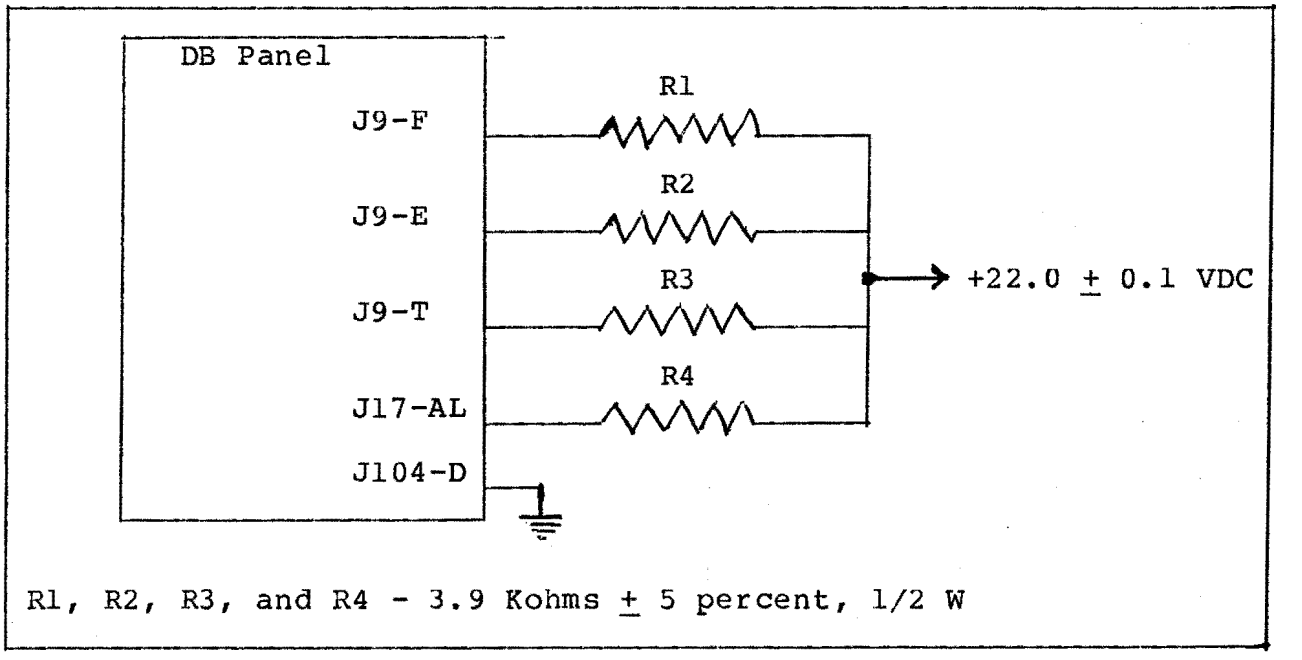


FIGURE 6. Distribution box test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.11 Indicators. With a potential of +22.0 ± 0.1 VDC applied between the pins specified in Table V and J104-D(-), the associated indicator lamp shall LIGHT. The K3 indicator lamp shall LIGHT when ground (J104-D) is applied to J17-AS.

TABLE V. Indicator lamp tests.

Applied Voltage Pin (+)	Indicator Lamp Lights
J13-i	K1
J17-AU	K2
J11-D	K4
J13-P	K17
J15-A	Gun Motor
J9-A	Azimuth Ampl No. 1
J9-k	Azimuth Ampl No. 2
J9-e	Elevation Ampl

3.4.2.12 Switch functions. The resistance between the sets of pins specified in Table VI shall be less than 0.5 ohms when the corresponding switch is in the position specified. In any other switch position, the resistance shall be greater than 1 megohm.

TABLE VI. Resistance test.

SWITCH	POSITION	PINS	
GUN POWER	ON	J17-AX	J17-AK
CLEAR MODE	MAN	J17-AE	J17-AK
CLEAR MODE	AUTO	J17-w	J17-AK
CLEAR MODE	AUTO	J17-x	J17-y
CLEAR MODE	MAN	J17-x	J17-z
TRIGGER	ON	J17-AJ	J17-AF
RATE, BURST LIMIT	HI,ON	J17-n	J17-r
RATE	LO	J17-p	J17-q
RATE	LO	J17-k	J17-m
RATE	HI	J17-j	J17-i
RATE	LO	J17-j	J17-p
LAST ROUND	OFF	J14-D	J14-G
LAST ROUND	ON	J14-D	J14-E
ACTION SWITCH	ON	J17-AD	J17-AC
SYSTEM POWER	ON	J17-AC	J17-AW
RADAR ELECTRONICS	ON	J17-BE	J17-BB
RIGHT LIMIT	ON	J17-AN	J9-R
LEFT LIMIT	ON	J17-AB	J13-M
EL LIMIT	ON	J17-s	J9-W
ACQ TIME	ON	J9-g	J13-a
LOAD	ON	J14-H	J14-C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

3.4.2.13 Test points. The resistance between the pins and test points specified in Table VII shall be less than 0.5 ohms when the RELAY TIME SELECTORS, CHANNEL A and CHANNEL B switches are in the position specified. In any other position, the resistance shall be greater than 1 megohm.

TABLE VII. Resistance test.

CHANNEL A POSITION	CHANNEL B POSITION	PIN(s)	TEST POINT
K5	K5	J13-V	TP6
K6		J13-U	
K7		J13-W	
K8		J13-d	
K9		J13-c	
K10		J13-P	
K11		J13-E	
K12		J13-D	
BRAKE		J13-p	
OFF	K5	J13-J	
K5	K6	J13-H	
K6		J13-k	
K7		J13-Y	
K8		J13-b	
K9		J13-s	
K10		J13-t	
K11		J13-q	
K12		J13-C	
BRAKE		J13-F	
OFF	K6	J13-n	
K5	K7	J13-r	
K6		J13-G	
K7		J13-f	
K8		J13-N	
K9		J13-j	
K10		J13-B	
K11		J13-h	
K12		J13-i	
BRAKE		J13-X	
OFF	K7	J13-A	
K5	K8	J13-Z	
K6		J13-e	
K7		J13-R	
K8		J13-L	
K9		J13-q	
K10		J13-T	
K11		J13-M	
K12		J13-a	
BRAKE		J13-K	
OFF	K8	J17-BM,BT	TP6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Resistance test. (Continued)

CHANNEL A POSITION	CHANNEL B POSITION	PIN(s)	TEST POINT
K5	K9	J17-BS,BU	TP6
K6	↑	J17-BA,BV	↑
K7	|	J17-AP,AZ	|
K8	|	J17-AR,BJ	|
K9	|	J17-AV	|
K10	|	J17-AX,AH	|
K11	|	J17-AD	|
K12	|	J17-AC	|
BRAKE	|	J17-AW	|
OFF	K9	J17-BE	|
K5	K10	J17-BB	|
K6	↑	J17-i	|
K7	|	J17-j	|
K8	|	J17-w	|
K9	|	J17-AE	|
K10	|	J17-y	|
K11	|	J17-z	|
K12	|	J17-x	|
BRAKE	|	J17-AK	|
OFF	K10	J17-m	|
K5	K11	J17-AU	|
K6	↑	J17-AS	|
K7	|	J17-q	|
K8	|	J17-p	|
K9	|	J17-AJ	|
K10	|	J17-AF	|
K11	|	J17-k	|
K12	|	J17-n	|
BRAKE	|	J17-r	|
OFF	K11	J17-AL	|
K5	K12	J17-AM	|
K6	↑	J17-AA	|
K7	|	J17-v	|
K8	|	J17-u	|
K9	|	J17-BR	|
K10	|	J17-t	|
K11	|	J17-g	|
K12	|	J17-d	|
BRAKE	|	J17-f	|
OFF	K12	J17-c	TP6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Resistance test. (Continued)

CHANNEL A POSITION	CHANNEL B POSITION	PIN(s)	TEST POINT
K5	BRAKE	J16-E,F	TP6
K6	↑	J16-C,G	↑
K7	|	J16-K,J10-A	|
K8	|	J12-A,B	|
K9	|	J12-D,E	|
K10	|	J15-K	|
K11	|	J15-D	|
K12	|	J15-F	|
BRAKE	|	J15-B	|
OFF	BRAKE	J15-A	|
K5	OFF	J14-G	|
K6	↑	J14-D	|
K7	|	J14-E	|
K8	|	J14-A,	|
	|	J11-B	|
K9	|	J14-L	|
K10	|	J14-N	|
K11	|	J14-B,F	|
K12	|	J14-H	|
BRAKE	|	J14-C	|
OFF	OFF	J14-J	TP6
	K5	J14-P	TP7
	K6	J11-A,H	↑
	K7	J11-E	|
	K8	J11-D	|
	K9	J9-e	|
	K10	J9-B	|
	K11	J9-T	|
	K12	J9-Z	|
	BRAKE	J9-k	|
	OFF	J9-A	TP7
	K5	J9-E	TP8
	K6	J9-F	↑
	K7	J9-j	|
	K8	J9-Y	|
	K9	J9-n	|
	K10	J9-r	|
	K11	J9-f	|
	K12	J9-V	|
	BRAKE	J9-c	|
	OFF	J9-S	TP8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Resistance test. (Continued)

CHANNEL A POSITION	CHANNEL B POSITION	PIN(s)	TEST POINT
	K5	J9-C	TP9
	K6	J9-h	
	K7	J9-s	
	K8	J9-W	
	K9	J9-i,G	
	K10	J9-R	
	K11	J9-P	TP9

3.4.2.14 5.0 Amperes circuit breaker. The circuit breaker shall open when subjected to a current greater than 5.0 amperes. The maximum overload shall never be greater than 200 percent of the circuit breaker's rated value. At the maximum overload condition, the circuit breaker shall open within 60 seconds.

3.5 First article. Unless otherwise specified in the contract or purchase order, a sample should be subjected to first article inspection (see 4.2 and 6.2).

3.6 Workmanship. The requirements of MIL-STD-454, MIL-F-13926 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article (initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be one test set manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.3.2 Examinations and tests. Except as otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Table VIII and IX and method of inspection in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.3.2.1 Components and subassemblies. Printed circuit subassemblies, shall be inspected in accordance with the inspection provisions of MIL-C-60898. All other components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

4.3.2.2 Control tests. One test set shall be selected at random as a control sample from each 100 produced or from each months production, whichever occurs first. The control samples shall meet the requirements and tests of Table VIII after they have met the requirements and tests of Table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE VIII. Test plan requirements.

TEST	REQUIREMENT	TEST PROCEDURES
Fabrication	3.1	Applicable drawings-visual
General specifications	3.2	Visual-tactile
Vibration	3.3.1	4.5.1.2
Temperature	3.3.2	4.5.1.3
Humidity	3.3.3	4.5.1.4
Shock	3.3.4	4.5.1.5
Altitude	3.3.5	4.5.1.6

4.3.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Functional tests. The requirements and tests in Table IX shall be inspected on a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50° to +90°F

b. Relative humidity 90% RH max

c. Pressure 28 to 32 inches of Hg

TABLE IX. Functional test. (100 percent inspection)

Characteristic	Requirement	Procedure
101 Continuity test	3.4.1	4.5.2.1
102 Conditions of Operation	3.4.2.1	4.5.3.1
103 Test System Power Switches	3.4.2.2	4.5.3.2
104 K12 Lamp delay	3.4.2.3	4.5.3.3
105 K6 Lamp delay	3.4.2.4	4.5.3.4
106 K8 Lamp delay	3.4.2.5	4.5.3.5
107 K11 Lamp delay	3.4.2.6	4.5.3.6
108 Low Input	3.4.2.7.1	4.5.3.7.1
109 High Input	3.4.2.7.2	4.5.3.7.2
110 K7 Lamp delay	3.4.2.8	4.5.3.8
111 K9 Lamp delay	3.4.2.9	4.5.3.9
112 Switch Indicators	3.4.2.10	4.5.3.10
113 Indicators	3.4.2.11	4.5.3.11
114 Switch Functions	3.4.2.12	4.5.3.12
115 Test Points	3.4.2.13	4.5.3.13
116 Circuit Breaker	3.4.2.14	4.5.3.14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements cited in the contract. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall provide inspection equipment compatible with the "Methods of inspection" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Inspection equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

TABLE X. Inspection equipment list.

Item	Description
1. POWER SUPPLY (2 ea.)	Nominal value of plus 5 vdc; 0-30 vdc, adjustable voltage. Load regulation 1% or less from 10 to 500 milliamperes. Less than 0.5% ripple.
2. POWER SUPPLY (2 ea.)	Nominal value of plus 22 vdc; 0-30 vdc, adjustable voltage. Load regulation 0.5% or less from 250 milliamperes to 5 amperes. Less than 0.5% ripple.
3. DIGITAL VOLT-OHMMETER	Automatic scaling; resistance accuracy and resolution of 0.2% or better from 1 to 1000 kilohms. DC voltage accuracy and resolution of 0.1% or better from 1 to 100 vdc. Input impedance greater than 10 megohms on all voltage scales.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

TABLE X. Inspection equipment list. (Continued)

Item		Description
4.	OSCILLOSCOPE	Dual channel vertical input; input impedance 1 megohm or greater. Rise time less than 2 microseconds.
5.	POWER SUPPLY	0-30 vdc, adjustable voltage, 600 milliampere. Less than 1% ripple.
6.	ASSORTED COMPONENTS	Resistor (4 ea.) 3.9 kilohms ±5%, 1/2 watt. Clip leads with alligator clips.
7.	POWER SUPPLY	A D.C. power supply and an ammeter indicating load current and an adjustable current limiting control. The capacity will be 75 amperes or greater.

4.5 Methods of inspection.

4.5.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.5.1.2 Vibration. The test set shall be mounted on the vibration table in normal service position. The test shall be performed as specified in Method 514, Procedure I, Category 1, of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration the test set shall then be tested as given in the sub-paragraphs of 4.5.2 and 4.5.3 for compliance with 3.3.1.

4.5.1.3 Temperature.

4.5.1.3.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501, of MIL-STD-810 with constant temperature extremes as specified in 3.3.2. The operational checkout shall be as specified in 4.5.2 and 4.5.3 for compliance with 3.3.2.

4.5.1.3.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.2. The operational checkout shall be as specified in 4.5.2 and 4.5.3 for compliance with 3.3.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.5.1.4 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only and at standard ambient conditions. The operational checkout shall be as specified in 4.5.2 and 4.5.3 for compliance with 3.3.3.

4.5.1.5 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be specified in 3.3.4. After the shock impulses, the test set shall be tested as specified in 4.5.2 and 4.5.3 for compliance with 3.3.4.

4.5.1.6 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.5.2 and 4.5.3 for compliance with 3.3.5.

4.5.2 Electrical test.

4.5.2.1 Continuity test. A continuity test shall be performed using an ohmmeter and visual examination to verify compliance with 3.4.1.

4.5.3 Functional.

4.5.3.1 Conditions of operation. All tests shall be conducted with the power supplies (Items 1 and 2) connected as shown in Figure 1, and with the TEST SYSTEM POWER +5 VDC and +24 VDC switches in the "ON" position. Unless specified, all subsequent tests will be conducted with these conditions of operation.

4.5.3.2 Test system power switches. Verify the +5 VDC and +24 VDC indicator lamp conditions with the TEST SYSTEM POWER switches in their "ON/OFF" positions as outlined in Table XI.

TABLE XI. Power switches tests.

TEST SYSTEM POWER SWITCHES		INDICATOR LAMPS	
+5 VDC ON/OFF	+24 VDC ON/OFF	+5 VDC	+24 VDC
OFF	OFF	OFF	OFF
ON	OFF	ON	OFF
ON	ON	ON	ON
OFF	ON	OFF	OFF

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.5.3.3 K12 LAMP delay. Connect oscilloscope (Item 4 of 4.4.1.1) to CHANNEL A and CHANNEL B connectors of DB panel and verify the requirements of 3.4.2.3. The presentation of the oscilloscope shall be as shown in Figure 7.

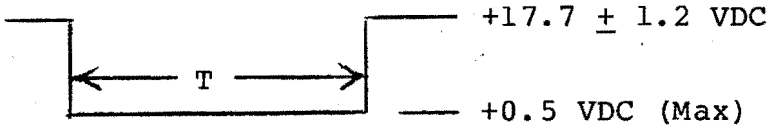


Figure 7. K12 Output pulse.

1. T for Step e. shall be 240 ± 5 msec.
2. T for Step g. shall be 160 ± 5 msec.
3. Step i., the output shall remain at +0.5 VDC max.

4.5.3.4 K6 Lamp delay. Connect oscilloscope (Item 4) to CHANNEL A and CHANNEL B connectors and verify the requirements of 3.4.2.4. The presentation on the oscilloscope shall be as shown in Figure 8.

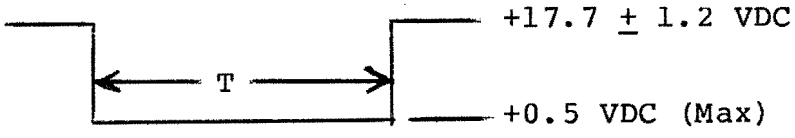


Figure 8. K6 Output pulse.

1. T for Step g. shall be 540 ± 5 msec.
2. T for Step i. shall be 460 ± 5 msec.
3. For Step k, the output shall remain at +0.5 VDC max.

4.5.3.5 K8 Lamp delay. Connect oscilloscope (Item 4) to CHANNEL A and CHANNEL B connectors and verify the requirements of 3.4.2.5. The presentation on the oscilloscope shall be as shown in Figure 9.

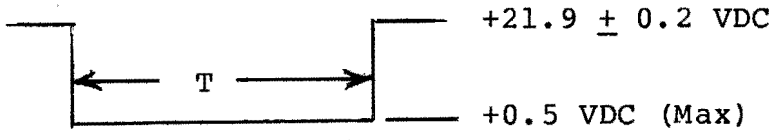


Figure 9. K8 Output pulse.

1. T for Step e. shall be 240 ± 5 msec.
2. T for Step i. shall be 160 ± 5 msec.
3. Voltage output for Step K shall be +2.9 ± 0.2 VDC.
4. Voltage output for Step l shall be a max of +0.5 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.5.3.6 K11 Lamp delay. Connect oscilloscope (Item 4) to CHANNEL A and CHANNEL B connectors and verify the requirements of 3.4.2.6. The presentation on the oscilloscope shall be as shown in Figure 10.

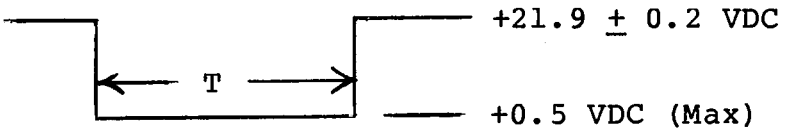


Figure 10. K11 Output pulse.

1. T for Step f. shall be 1065 ± 5 msec.
2. T for Step g. shall be 935 ± 5 msec.
3. Voltage output for Step h shall be +21.9 ± 0.2 VDC.

4.5.3.7 Relay time selectors.

4.5.3.7.1 Low input. Using the digital volt-ohmmeter (Item 3) measure the DC voltages between INPUT A and GRD also between INPUT B and GRD connectors and verify the requirements of 3.4.2.7.1.

4.5.3.7.2 High input. Using the digital volt-ohmmeter (Item 3) measure the DC voltages between INPUT A and GRD connectors also between INPUT B and GRD connectors and verify the requirements of 3.4.2.7.2.

4.5.3.8 K7 Lamp delay. Using the digital volt-ohmmeter (Item 3) measure the DC voltages between INPUT A and GRD connectors and also between INPUT B and GRD connectors and verify the requirements of 3.4.2.8.

4.5.3.9 K9 Lamp delay. Using the digital volt-ohmmeter (Item 3) measure the DC voltages between INPUT A and GRD connectors and also between INPUT B and GRD connectors and verify the requirements of 3.4.2.9.

4.5.3.10 Switch indicators. Referring to Table IV, operate the specified switches in the "ON" position and then in the "OFF" position and verify that the associated indicator lamp response conforms to the requirements of 3.4.2.10.

4.5.3.11 Indicators. Apply +22.0 ± 0.1 VDC (Item 2 POWER SUPPLY) to the pins as listed in Table V and verify the Indicator lamps meet the requirements of 3.4.2.11.

4.5.3.12 Switch functions. Disconnect all power supply connectors from DB panel and with volt-ohmmeter (Item 3) check for continuity between the sets of pins as listed in Table VI and verify the requirements of 3.4.2.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

4.5.3.13 Test points. Disconnect all power supply connectors from DB panel and with volt-ohmmeter (Item 3) check for continuity between the pin(s) and test point when the RELAY TIME SELECTORS CHANNEL A and CHANNEL B switches are in the various position as outlined in Table VII and verify the requirements of 3.4.2.13.

4.5.3.14 5 Amperes circuit breaker. Disconnect +5 vdc power supply from J104-F. Connect power supply (Item 1) across the terminals of 5 amperes circuit breaker and adjust current to 10 amperes. Circuit breakers shall open in 60 seconds. Remove power supply and reset circuit breaker.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 10554006.

6. NOTES

6.1 Intended use. This test set is designed for rack-mounting into the AN/TSM-100A Electronics System Test Set, which is used for trouble shooting and performance checking of specific weapon system components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each (see 6.3).

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.4 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A218)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.